Urban Environmental Intelligence API

API Reference Documentation • Version 1.0

Base URL
https://urban-env-api.onrender.com

GitHub Repository
github.com/Heeeeeeliang/urban-env-api

Interactive Docs (Swagger UI)
https://urban-env-api.onrender.com/docs

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1. Overview

The Urban Environmental Intelligence API is a production-grade RESTful service that integrates real-time air quality measurements, weather observations, and AI-powered natural language analysis. It continuously ingests data from OpenAQ (PM2.5, PM10, NO2, O3) and Open-Meteo (temperature, humidity, wind speed, pressure), validates every reading, and stores results in PostgreSQL for time-series analytics.

The API exposes five analytical query patterns — trend analysis, multi-city comparison, anomaly detection, cross-metric correlation, and city ranking — all backed by PostgreSQL-optimised SQL with composite indexes. An AI insight endpoint sends pre-aggregated statistics to Anthropic Claude, which generates natural language environmental briefings in plain English. The system also exposes an MCP (Model Context Protocol) server, enabling Claude Desktop and other MCP-compatible clients to query environmental data through native tool integrations.

1.1 Base URL & Versioning

All endpoints are prefixed with /api/v1. The API is deployed on Render's free tier.

```
Base URL:    https://urban-env-api.onrender.com
API prefix:  /api/v1
Swagger UI:  https://urban-env-api.onrender.com/docs
ReDoc:       https://urban-env-api.onrender.com/redoc
Health:      https://urban-env-api.onrender.com/health
```

Note: The service runs on Render's free tier. The first request after a period of inactivity may take up to 30 seconds while the instance cold-starts.

1.2 Authentication

Most endpoints require an API key passed as the X-Api-Key request header. The AI insight endpoint (/api/v1/ai/insight/{city_id}) and the health endpoint are publicly accessible without authentication.

```
X-Api-Key: your-api-key
```

API keys support three permission tiers:

- read — GET endpoints only
- write — GET + POST + PUT + DELETE
- admin — all endpoints including administrative operations

Parameter	Type	Required	Description
X-Api-Key	string	See note	API key in request header. Required for all endpoints except /health and /api/v1/ai/insight/{city_id}.

1.3 Response Format

All responses are JSON (application/json). Successful responses return the resource or a list. Error responses follow a consistent envelope:

```
{
  "detail": "Human-readable error message"
}
```

1.4 HTTP Status Codes

Code	Meaning
`200 OK`	Request successful. Response body contains the resource or collection.
`201 Created`	Resource successfully created (POST). Response body contains the new resource.
`204 No Content`	Soft-delete successful (DELETE). No response body.
`400 Bad Request`	Invalid request parameters or body. Check the detail field.
`401 Unauthorized`	Missing or unrecognised X-Api-Key header.
`403 Forbidden`	API key recognised but insufficient permissions for this operation.
`404 Not Found`	The requested city_id or resource does not exist.
`422 Unprocessable Entity`	FastAPI validation error — required field missing or wrong type.
`429 Too Many Requests`	Rate limit exceeded. Default: 60 GET req/min, 20 POST req/min per IP.
`503 Service Unavailable`	AI service unavailable and no cached insight exists.

1.5 Rate Limiting

Rate limiting is enforced per client IP using slowapi. Limits apply per minute:

- GET requests: 60 per minute per IP
- POST requests: 20 per minute per IP

When the limit is exceeded, the API returns HTTP 429 with a Retry-After header indicating when the limit resets.

2. Cities Endpoints

The Cities resource represents geographic locations monitored for air quality and weather data. Each city is identified by a slug (e.g., london, manchester). Cities support full CRUD operations. DELETE is a soft-delete — the city's is_active flag is set to false and it is excluded from analytics, but historical readings are preserved.

2.1 Endpoint Summary

Method	Path	Description	Auth
GET	`/api/v1/cities`	List all active cities (paginated)	X-Api-Key
POST	`/api/v1/cities`	Register a new city for monitoring	X-Api-Key
GET	`/api/v1/cities/{city_id}`	Get city details by slug	X-Api-Key
PUT	`/api/v1/cities/{city_id}`	Update city metadata (partial update)	X-Api-Key
DELETE	`/api/v1/cities/{city_id}`	Soft-delete a city (sets is_active=false)	X-Api-Key

2.2 List Cities — GET /api/v1/cities

Returns a paginated list of all active cities. Inactive (soft-deleted) cities are excluded.

Query Parameters

Parameter	Type	Required	Description
skip	integer	No	Number of records to skip for pagination. Default: 0. Minimum: 0.
limit	integer	No	Maximum records to return. Default: 100. Range: 1–100.

Example Request

```
curl -H "X-Api-Key: your-key" \
  "https://urban-env-api.onrender.com/api/v1/cities?skip=0&limit=10"
```

Example Response — 200 OK

```
[
  {
    "id": "london",
    "name": "London",
    "country": "United Kingdom",
    "country_code": "GB",
    "latitude": 51.5074,
    "longitude": -0.1278,
    "timezone": "Europe/London",
    "is_active": true,
    "created_at": "2026-02-27T10:00:00Z",
    "updated_at": "2026-02-27T10:00:00Z"
  }
]
```

2.3 Create City — POST /api/v1/cities

Registers a new city for monitoring. The city slug (id) is auto-generated from the city name (lowercased, spaces replaced with hyphens). The background ingestion pipeline will begin fetching data for this city on its next scheduled run (default: every 60 minutes).

Request Body (application/json)

Parameter	Type	Required	Description
name	string	**Yes**	City display name (e.g., "Manchester"). Used to derive the id slug.
country	string	**Yes**	Full country name (e.g., "United Kingdom").
country_code	string	**Yes**	ISO 3166-1 alpha-2 country code (e.g., "GB"). 2 characters.
latitude	float	**Yes**	Geographic latitude in decimal degrees. Range: -90 to 90.
longitude	float	**Yes**	Geographic longitude in decimal degrees. Range: -180 to 180.
timezone	string	**Yes**	IANA timezone identifier (e.g., "Europe/London").

Example Request

```
curl -X POST "https://urban-env-api.onrender.com/api/v1/cities" \
  -H "Content-Type: application/json" \
  -H "X-Api-Key: your-key" \
  -d '{
    "name": "Manchester",
    "country": "United Kingdom",
    "country_code": "GB",
    "latitude": 53.4808,
    "longitude": -2.2426,
    "timezone": "Europe/London"
  }'
```

Example Response — 201 Created

```
{
  "id": "manchester",
  "name": "Manchester",
  "country": "United Kingdom",
  "country_code": "GB",
  "latitude": 53.4808,
  "longitude": -2.2426,
  "timezone": "Europe/London",
  "is_active": true,
  "created_at": "2026-03-01T12:00:00Z",
  "updated_at": "2026-03-01T12:00:00Z"
}
```

2.4 Get City — GET /api/v1/cities/{city_id}

Returns full details for a single city by its slug identifier.

Path Parameters

Parameter	Type	Required	Description
`city_id`	`string`	**Yes**	City slug identifier (e.g., london, manchester, beijing).

Example Request

```
curl -H "X-Api-Key: your-key" \
  "https://urban-env-api.onrender.com/api/v1/cities/london"
```

Errors

- 404 Not Found — city_id does not exist or the city has been soft-deleted.

2.5 Update City — PUT /api/v1/cities/{city_id}

Partially updates city metadata. All fields are optional — only provided fields are updated. The city id (slug) cannot be changed.

Request Body (application/json) — all fields optional

Parameter	Type	Required	Description
`name`	`string`	No	New display name.
`country`	`string`	No	New country name.
`country_code`	`string`	No	New ISO 3166-1 alpha-2 country code.
`latitude`	`float`	No	New latitude. Range: -90 to 90.
`longitude`	`float`	No	New longitude. Range: -180 to 180.
`timezone`	`string`	No	New IANA timezone identifier.

Example Response — 200 OK

Returns the full updated city object (same schema as Create City response).

2.6 Delete City — DELETE /api/v1/cities/{city_id}

Soft-deletes a city by setting is_active = false. The city is immediately excluded from all analytics queries and the ingestion pipeline. Historical readings are preserved in the database and are not deleted.

Example Response — 204 No Content

Empty response body on success.

Errors 404 Not Found — city_id does not exist or is already inactive.

3. Analytics Endpoints

The Analytics resource provides five PostgreSQL-optimised query patterns over historical readings. All analytics endpoints require X-Api-Key authentication and accept a city_id path parameter. Analytics are computed on-the-fly from raw readings using window functions, z-score calculations, and Pearson correlation — no pre-aggregated cache is used (unlike AI insights).

3.1 Endpoint Summary

Method	Path	Description	Auth
GET	`/api/v1/analytics/trend/{city_id}`	Time-aggregated trend (hourly/daily/weekly)	X-Api-Key
GET	`/api/v1/analytics/compare`	Multi-city comparison with % change	X-Api-Key
GET	`/api/v1/analytics/anomalies/{city_id}`	Z-score anomaly detection	X-Api-Key
GET	`/api/v1/analytics/correlation/{city_id}`	Pearson correlation between two parameters	X-Api-Key
GET	`/api/v1/analytics/ranking`	Rank cities by average parameter value	X-Api-Key

3.2 Trend Analysis — GET /api/v1/analytics/trend/{city_id}

Returns time-aggregated averages, minimums, and maximums for a chosen parameter and city, grouped by configurable intervals (hourly, daily, or weekly). Uses PostgreSQL date_trunc for efficient bucketing with composite index support.

Path Parameters

Parameter	Type	Required	Description
city_id	string	Yes	City slug (e.g., london).

Query Parameters

Parameter	Type	Required	Description
parameter	string	Yes	Measurement parameter. One of: pm25, pm10, no2, o3, temperature, humidity, wind_speed, pressure.
days	integer	No	Number of days to look back. Default: 7. Range: 1–90.
interval	string	No	Aggregation interval. One of: hour, day, week. Default: day.

Example Request

```
curl -H "X-Api-Key: your-key" \
```

```
"https://urban-env-api.onrender.com/api/v1/analytics/trend/london?parameter=pm25&days=7&interval=day"
```

Example Response — 200 OK

```
{
  "city_id": "london",
  "city_name": "London",
  "parameter": "pm25",
  "unit": "ug/m3",
  "interval": "day",
  "data": [
    {
      "period": "2026-02-20T00:00:00Z",
      "avg": 12.3,
      "min": 8.1,
      "max": 18.7,
      "reading_count": 24
    },
    {
      "period": "2026-02-21T00:00:00Z",
      "avg": 14.1,
      "min": 9.2,
      "max": 45.0,
      "reading_count": 24
    }
  ],
  "total_readings": 168
}
```

3.3 City Comparison — GET /api/v1/analytics/compare

Compares multiple cities side-by-side for a given parameter over a specified time window, returning average, minimum, maximum, and percentage change relative to a rolling baseline. Useful for identifying which cities have the best or worst air quality for a given pollutant.

Query Parameters

Parameter	Type	Required	Description
city_ids	string[]	**Yes**	Comma-separated list of city slugs to compare (e.g., london,manchester,beijing). Minimum 2 cities.
parameter	string	**Yes**	Measurement parameter. Same options as trend endpoint.
days	integer	No	Comparison window in days. Default: 7. Range: 1–30.

Example Request

```
curl -H "X-Api-Key: your-key" \

"https://urban-env-api.onrender.com/api/v1/analytics/compare?city_ids=london,manchester&parameter=pm25&days=7"
```

Example Response — 200 OK

```
{
  "parameter": "pm25",
  "unit": "ug/m3",
  "period_days": 7,
  "comparisons": [
    {
      "city_id": "london",
      "city_name": "London",
      "avg": 12.3,
      "min": 8.1,
      "max": 45.0,
      "reading_count": 168,
      "pct_change": -5.2
    },
    {
      "city_id": "manchester",
      "city_name": "Manchester",
      "avg": 15.7,
      "min": 9.0,
      "max": 38.2,
      "reading_count": 162,
      "pct_change": 8.1
    }
  ]
}
```

3.4 Anomaly Detection — GET /api/v1/analytics/anomalies/{city_id}

Identifies statistically unusual readings using z-score analysis. For each reading, the pipeline computes a rolling 30-day average and standard deviation; readings whose z-score exceeds the sensitivity threshold are flagged as anomalies. Anomalies are classified by severity based on their z-score magnitude.

Path Parameters

Parameter	Type	Required	Description
`city_id`	`string`	**Yes**	City slug.

Query Parameters

Parameter	Type	Required	Description
`parameter`	`string`	**Yes**	Measurement parameter to analyse.
`days`	`integer`	No	Lookback window in days. Default: 30. Range: 7–90.
`sensitivity`	`float`	No	Z-score threshold. Lower = more sensitive. Default: 2.0. Range: 1.0–5.0.

Severity Classification

Severity	Z-Score Range	Interpretation
low	`sensitivity ≤ z < sensitivity + 1`	Mildly unusual — worth monitoring

Severity	Z-Score Range	Interpretation
medium	`sensitivity+1 ≤ z < sensitivity+2`	Notably unusual — likely a real event
high	`z ≥ sensitivity + 2`	Strongly anomalous — pollution spike or sensor error

Example Response — 200 OK

```
{
  "city_id": "london",
  "city_name": "London",
  "parameter": "pm25",
  "unit": "ug/m3",
  "sensitivity": 2.0,
  "anomalies": [
    {
      "timestamp": "2026-02-20T19:00:00Z",
      "value": 45.0,
      "rolling_avg": 12.5,
      "z_score": 17.4,
      "severity": "high"
    }
  ],
  "total_readings_analysed": 720
}
```

3.5 Correlation Analysis — GET /api/v1/analytics/correlation/{city_id}

Computes the Pearson correlation coefficient between two parameters using the SQL corr() function. A value close to +1 indicates a strong positive correlation; close to -1 indicates strong negative correlation; near 0 indicates no linear relationship. Useful for exploring relationships such as PM2.5 vs temperature or humidity vs wind speed.

Path Parameters

Parameter	Type	Required	Description
`city_id`	`string`	**Yes**	City slug.

Query Parameters

Parameter	Type	Required	Description
`parameter_x`	`string`	**Yes**	First parameter (x-axis).
`parameter_y`	`string`	**Yes**	Second parameter (y-axis).
`days`	`integer`	No	Lookback window in days. Default: 30. Range: 7–90.

Example Request

```
curl -H "X-Api-Key: your-key" \
```

```
"https://urban-env-api.onrender.com/api/v1/analytics/correlation/london?parameter_x
=pm25&parameter_y=temperature&days=30"
```

Example Response — 200 OK

```
{
  "city_id": "london",
  "parameter_x": "pm25",
  "parameter_y": "temperature",
  "pearson_r": -0.62,
  "sample_size": 694,
  "interpretation": "Moderate negative correlation: higher temperatures are
associated with lower PM2.5 levels.",
  "period_days": 30
}
```

3.6 City Ranking — GET /api/v1/analytics/ranking

Ranks all active cities by their average value for a given parameter over a specified time window. Cities with fewer readings than the minimum sample threshold are excluded to avoid ranking noise from sparse data.

Query Parameters

Parameter	Type	Required	Description
parameter	string	**Yes**	Parameter to rank by.
days	integer	No	Lookback window in days. Default: 7. Range: 1–30.
order	string	No	Sort order. asc (best air quality first) or desc (worst first). Default: asc.
min_readings	integer	No	Minimum reading count threshold. Cities below this are excluded. Default: 10.

Example Response — 200 OK

```
{
  "parameter": "pm25",
  "unit": "ug/m3",
  "period_days": 7,
  "order": "asc",
  "rankings": [
    { "rank": 1, "city_id": "reykjavik", "city_name": "Reykjavik", "avg": 3.2,
"reading_count": 168 },
    { "rank": 2, "city_id": "london",    "city_name": "London",    "avg": 12.3,
"reading_count": 168 },
    { "rank": 3, "city_id": "beijing",   "city_name": "Beijing",   "avg": 58.7,
"reading_count": 166 }
  ]
}
```

4. AI Insight Endpoint

The AI insight endpoint demonstrates the highest-tier GenAI integration in the architecture: using an LLM not as a code generator, but as a runtime component of the API that produces natural language environmental briefings from pre-computed analytics.

The pipeline separates deterministic analytics (PostgreSQL) from probabilistic narration (LLM). The AI never performs arithmetic — it interprets patterns that have already been computed by the database. This architectural choice ensures that even if the LLM produces an inaccurate narrative, the underlying numeric statistics remain correct and verifiable by API consumers.

4.1 Get AI Insight — GET /api/v1/ai/insight/{city_id}

Produces a natural language briefing of a city's environmental conditions over the last 7 days, powered by Anthropic Claude.

Path Parameters

Parameter	Type	Required	Description
`city_id`	`string`	**Yes**	City slug (e.g., london).

Query Parameters

Parameter	Type	Required	Description
`include_context`	`boolean`	No	Include the pre-aggregated data payload in the response body. Default: true. Set to false to reduce response size.

Authentication

This endpoint is publicly accessible. No X-Api-Key header is required.

Caching Behaviour

- Cache HIT: ~5ms response time (database lookup only)
- Cache MISS: 3–8 seconds (full analytics pipeline + LLM API call)
- Cache FAIL: Expired cache served with X-Cache-Status: stale warning header

Cache TTL aligns with the ingestion interval (default 3600 seconds). LLM costs scale with cities × ingestion frequency, not with request volume.

Pipeline Steps

- 1. Pre-aggregate statistics per parameter (SQL: AVG, MIN, MAX, trend)
- 2. Run anomaly detection via z-score analysis
- 3. Extract raw context window around each anomaly (±3 hours)
- 4. Compute cross-metric correlations (PM2.5 vs temperature, humidity)
- 5. Check cache for valid (non-expired) insight

- 6. If cache miss: assemble structured context → call Anthropic Claude API → cache result
- 7. Return insight with metadata including token usage

Example Request

```
curl "https://urban-env-api.onrender.com/api/v1/ai/insight/london"
```

Example Response — 200 OK


```
{
  "city_id": "london",
  "city_name": "London",
  "insight": "London's air quality showed a mixed picture over the past week. PM2.5
levels averaged 12.3 ug/m3, within the WHO annual guideline of 15 ug/m3, though a
notable spike to 45.0 ug/m3 on 20 February — a z-score of 17.4 — indicates a
localised pollution event, possibly linked to traffic congestion or industrial
activity during colder weather. The moderate negative correlation (-0.62) between
PM2.5 and temperature suggests that colder days are associated with worse air
quality, consistent with increased heating activity and reduced atmospheric
dispersion...",
  "generated_at": "2026-03-01T12:00:00Z",
  "expires_at": "2026-03-01T13:00:00Z",
  "cached": false,
  "model_used": "claude-haiku-4-5-20251001",
  "input_tokens": 847,
  "output_tokens": 312,
  "data_summary": {
    "parameters_analysed": ["pm25", "pm10", "no2", "o3", "temperature",
"humidity"],
    "anomaly_count": 1,
    "readings_analysed": 847
  }
}
```


Error Responses

- 404 Not Found — city_id does not exist.
- 503 Service Unavailable — Anthropic API is unreachable and no cached insight exists.

5. Health Endpoint

The health endpoint provides a real-time view of the service's operational status, including database connectivity and data freshness.

5.1 GET /health

Returns service health status. No authentication required. This endpoint is designed for infrastructure monitoring and uptime checks.

Example Response — 200 OK (Healthy)

```
{
  "status": "healthy",
  "database": "connected",
  "active_cities": 8,
  "total_readings": 14823,
  "last_ingestion": "2026-03-01T11:58:42Z",
  "ingestion_age_minutes": 1.3
}
```

Response Fields

Field	Type	Nullable	Description
status	string	No	Overall health. healthy or degraded.
database	string	No	Database connectivity. connected or disconnected.
active_cities	integer	No	Number of cities with is_active = true.
total_readings	integer	No	Total readings stored in the database.
last_ingestion	datetime	Yes	ISO 8601 timestamp of the last successful ingestion run. Null if no ingestion has occurred yet.
ingestion_age_minutes	float	Yes	Minutes since the last ingestion. Null if no ingestion has occurred yet.

6. Data Models

6.1 City

Field	Type	Nullable	Description
`id`	`string`	No	Slug identifier, auto-derived from name (e.g., london, new-york).
`name`	`string`	No	Display name (e.g., London).
`country`	`string`	No	Full country name.
`country_code`	`string`	No	ISO 3166-1 alpha-2 code (e.g., GB).
`latitude`	`float`	No	Geographic latitude, decimal degrees.
`longitude`	`float`	No	Geographic longitude, decimal degrees.
`timezone`	`string`	No	IANA timezone (e.g., Europe/London).
`is_active`	`boolean`	No	False if soft-deleted via DELETE endpoint.
`created_at`	`datetime`	No	UTC timestamp of city registration.
`updated_at`	`datetime`	No	UTC timestamp of last metadata update.

6.2 Reading (internal — not directly exposed)

Field	Type	Nullable	Description
`id`	`integer`	No	Auto-incremented primary key.
`city_id`	`string`	No	Foreign key to City.id.
`parameter`	`string`	No	Measurement type (pm25, temperature, etc.).
`value`	`float`	No	Normalised measurement value.
`raw_value`	`float`	Yes	Original API value before unit normalisation.
`unit`	`string`	No	Canonical unit (ug/m3 for pollutants, C for temperature).
`quality_flag`	`string`	No	Data quality: valid, suspect, or missing.
`timestamp`	`datetime`	No	UTC timestamp of the reading.
`source`	`string`	No	Data source: openaq or open_meteo.

6.3 Supported Parameters

Parameter	Unit	Source	Physical Bounds
`pm25`	`ug/m3`	OpenAQ	0 – 500 ug/m3
`pm10`	`ug/m3`	OpenAQ	0 – 1000 ug/m3
`no2`	`ug/m3`	OpenAQ	0 – 1000 ug/m3
`o3`	`ug/m3`	OpenAQ	0 – 500 ug/m3
`temperature`	`°C`	Open-Meteo	-90 – 60 °C

Parameter	Unit	Source	Physical Bounds
`humidity`	`%`	Open-Meteo	0 – 100 %
`wind_speed`	`km/h`	Open-Meteo	0 – 300 km/h
`pressure`	`hPa`	Open-Meteo	800 – 1100 hPa

7. MCP Integration

The API exposes a Model Context Protocol (MCP) server (mcp_server.py) that allows Claude Desktop, Cursor, and other MCP-compatible clients to query environmental data through native tool integrations — no HTTP required. Tools accept human-friendly city names, resolve them to slugs internally, and return formatted text optimised for LLM context windows.

7.1 Claude Desktop Configuration

```
{
  "mcpServers": {
    "urban-env": {
      "command": "python",
      "args": ["mcp_server.py"],
      "env": {
        "DATABASE_URL":
"postgresql+asyncpg://user:pass@localhost:5432/urban_env_db",
        "ANTHROPIC_API_KEY": "sk-ant-..."
      }
    }
  }
}
```

7.2 Available MCP Tools

Tool Name	Description
`get_city_air_quality`	Returns air quality status and 7-day trend for a city (PM2.5, PM10, NO2, O3) plus current weather conditions.
`compare_cities_pollution`	Side-by-side comparison of pollution levels across multiple cities with rankings. Accepts comma-separated city names.
`detect_pollution_anomalies`	Runs z-score analysis to find statistically unusual pollution events in the last 30 days.
`get_ai_environmental_insight`	Returns Claude-generated natural language briefing of environmental conditions, served from cache where available.

8. Error Reference

All error responses follow the standard FastAPI envelope:

```
{
  "detail": "Human-readable error message"
}
```

HTTP Code	When It Occurs	Resolution
400	Invalid query parameters (e.g., negative skip value, parameter not in allowed list)	Check parameter spelling and ranges documented in this reference.
401	X-Api-Key header missing entirely	Add the X-Api-Key header to your request.
403	API key valid but insufficient permissions (e.g., using a read key for POST)	Use a key with write or admin permissions.
404	city_id does not exist or has been soft-deleted	Verify the city slug via GET /api/v1/cities.
422	FastAPI schema validation failure — required field missing or wrong type	Check request body against schema. Common: missing required field, wrong JSON type.
429	Rate limit exceeded (60 GET/min or 20 POST/min per IP)	Back off and retry. Check the Retry-After response header.
503	AI service (Anthropic) unavailable and no cached insight exists	Retry later. For automation, implement exponential backoff.